

- 1) Construa as GLC p/ as linguagens
- 2) faça uma árvore de derivação p/ algum exemplo de palavra
- 3) simplifique a gramática
- 4) faça a FNC
- 5) faça a FNG
- 6) faça a Ap a partir da FNG

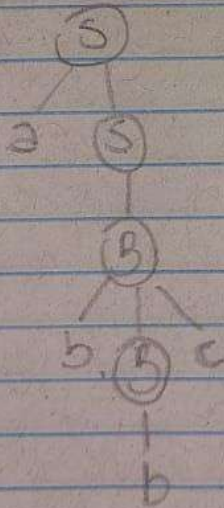
$$b) L = \{a^i b^j c^k \mid i > 0\}$$

1) GLC =

$$S \rightarrow aS \mid B$$

$$B \rightarrow bBc \mid b$$

2) Árvore de derivação $w = abbc$



3) Simplificar gramática

Não há o que simplificar

4) FNC

$$W \rightarrow a$$

$$X \rightarrow b$$

$$Y \rightarrow c$$

$$S \rightarrow WS \mid B$$

$$B \rightarrow XB Y \mid b$$

$$Z \rightarrow B Y \mid b$$

$$S \rightarrow WS \mid XZ$$

$$Z \rightarrow ZY \mid b$$

$$W \rightarrow a$$

$$X \rightarrow b$$

$$Y \rightarrow c$$

5) PFG

$S \rightarrow aS \mid B$

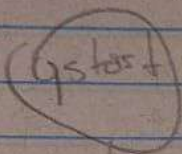
$B \rightarrow bBc \mid b$

$S \rightarrow aS \mid bXc \mid b$

$X \rightarrow bXc \mid b$

$C \rightarrow c$

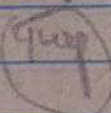
6)



$\epsilon, \epsilon \rightarrow \epsilon$



$\epsilon, \epsilon \rightarrow S$



$\epsilon, S \rightarrow S \quad \epsilon, \epsilon \rightarrow a$

$\epsilon, S \rightarrow C \quad \epsilon, \epsilon \rightarrow X \quad \epsilon, \epsilon \rightarrow b$

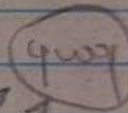
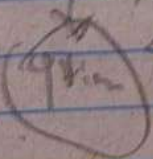
$\epsilon, S \rightarrow b$

$\epsilon, X \rightarrow C \quad \epsilon, \epsilon \rightarrow X \quad \epsilon, \epsilon \rightarrow b$

$\epsilon, X \rightarrow b$

$\epsilon, C \rightarrow c$

$\epsilon, \epsilon \rightarrow \epsilon$



$\epsilon, \epsilon \rightarrow \epsilon$
 $\epsilon, \epsilon \rightarrow \epsilon$
 $\epsilon, \epsilon \rightarrow \epsilon$
 $\epsilon, \epsilon \rightarrow \epsilon$



D S T Q Q S S

a) $L = \{a^i b^j c^{s+1} d^{2i} \mid i \geq 1, j \geq 1\}$

abccdd

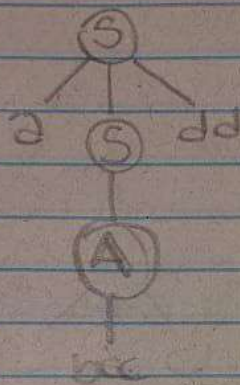
1) GLC

$$S \rightarrow aSd \mid A$$

$$A \rightarrow bAc \mid bcc$$

2) Árvore de derivação

$$w = abccdd$$



3) $S \rightarrow aSd \mid bSc \mid bcc$

4) FNC

$$w \rightarrow a$$

$$x \rightarrow b$$

$$y \rightarrow c$$

$$z \rightarrow d$$

$$S \rightarrow wSZZ \mid xSY \mid x\bar{Y}$$

$$H \rightarrow SY$$

$$i \rightarrow YY$$

$$j \rightarrow ZZ$$

$$w \rightarrow a$$

$$x \rightarrow b$$

$$y \rightarrow c$$

$$z \rightarrow d$$

$$S \rightarrow wM \mid xH \mid xi$$

$$H \rightarrow SY$$

$$i \rightarrow YY$$

$$j \rightarrow ZZ$$

$$w \rightarrow a$$

$$x \rightarrow b$$

$$y \rightarrow c$$

$$z \rightarrow d$$

$$M \rightarrow SJ$$

kajoma

D S T Q Q S S

5) FNG

$S \rightarrow aSAA \mid bSB \mid bBB$

$A \rightarrow d$

$B \rightarrow c$

6) Ap

